Double Pendulum and The Chaos Theory

AP Calculus EOC Project

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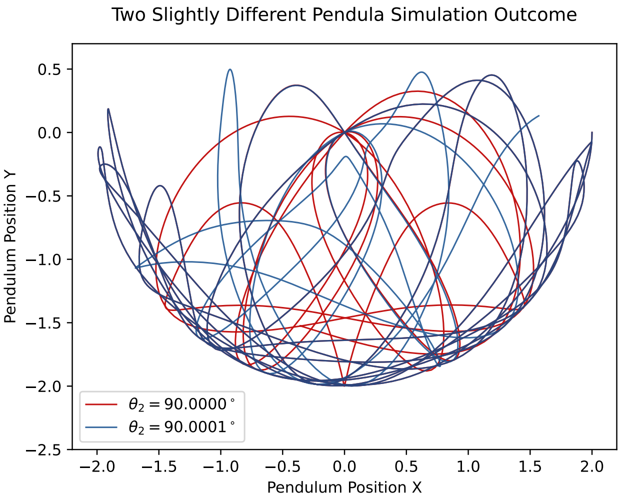
Related CALCAB Knowledge

Differential Equation: It is the core of simulation algorithm we applied in this project, the entire simulation was run based on a differential equation derived by *Lagrangian Mechanics*, so the differential equation we learned in unit 7 is the core of our project.

Limit and Approaching: In our project, the observation of the simulation accuracy as time step changes is also a core part. By speculating the result approaches a value as step decreases (becomes more accurate), we could choose a suitable time step for simulation that is both efficient and accurate.

Integration (by Part): For deriving Euler-Lagrange Equation.

Introduction To The Chaos Theory



Uncertain Outcome: In a chaotic system described in the Chaos Theory, even a *negligible* disturbance or initial differences could cause an incredibly different result.

Determinable, though Chaotic: Chaotic doesn't means random and indeterminate, instead, chaotic systems could be predicted by calculations and modeling if one have incredibly precise data, though the result is input-sensitive.

Introduction To DE Calculation Approaches

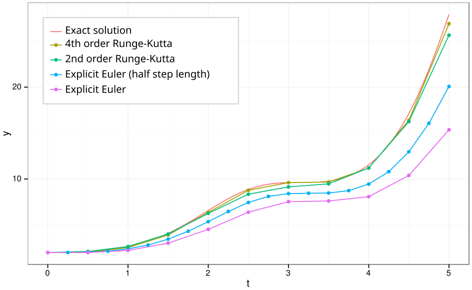


Figure: Comparison of the Runge-Kutta methods for the differential equation. Red line is the exact solution, 4th order Runge-Kutta is what we mainly use in this project, and is significantly more accurate. Figure from Wikimedia Commons, CC BY-SA 3.0

Fourth-Order Runge-Kutta (RK4) Approach:

Let $\dfrac{\mathrm{d}y}{\mathrm{d}x} = f(x, y)$,

$$y_{n+1} = y_n + \frac{\Delta x}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Where,

$$k_1 = f(x_n, y_n)$$

$$k_2 = f(x_n + \frac{\Delta x}{2}, y_n + \frac{\Delta x}{2} k_1)$$

$$k_3 = f(x_n + \frac{\Delta x}{2}, y_n + \frac{\Delta x}{2} k_2)$$

$$k_4 = f(x_n + \Delta x, y_n + \Delta x k_3)$$

Euler's Method: Simple DE Approximation

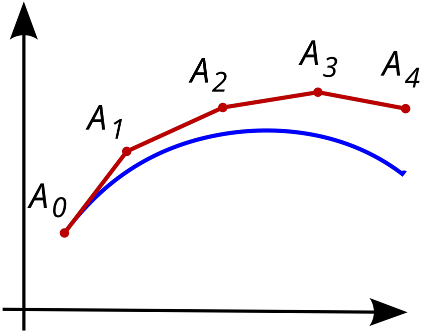


Figure: Illustration of the Euler method. The unknown curve is in blue, and its Euler approximation is in red. Figure from Wikimedia Commons, Public Domain.

Euler's Method:

$$\text{Let}\,\frac{\mathrm{d}y}{\mathrm{d}x} = f(x, y),$$
$$y_{n+1} = y_n + \Delta x f(x_n, y_n)$$

The key of Euler's Method is to use the slope at f_n as the approximation slope of the entire step (Δx), it's quite like a common sense lol. So Euler's Method underestimates when the actual function concaves up and overestimates when the function concaves down.

Double Pendulum, Where Newtonian Physics Dies

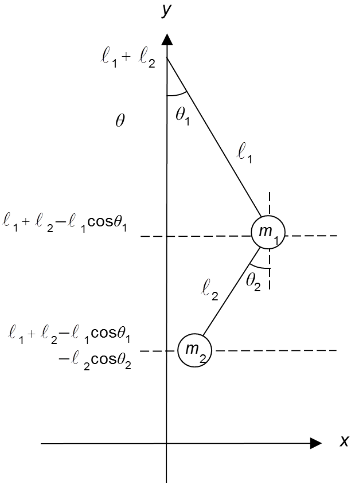


Figure from Wikimedia Commons account F=q(E+v^B), CC BY-SA 3.0

- Newtonian Physics is based on calculation of forces, and the motion of objects is determined by classic furmula $F = ma$.
- But in this problem, each variable are *interdependent* and *non-linear*, which makes this problem extremely complex (as complicated as three body problem) using Newtonian Physics to analyze.
- An approch physicists come up with is *Lagrangian Mechanics*, besically using Energy analysis to evaluate, and construct the system of differential equations.

Lagrangian Physics and Double Pendulum

Start from the Euler-Lagrangian Equation,

$$\frac{\partial L}{\partial q} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}} = 0 \tag{1}$$

where the Lagrangian $L = T - V$, ∂ (in multivariable calculus) basically means what d represents in normal calculus, q is the generalized coordinates, which is angle (θ_i) in our problem.
Citing 1, for double pendulums, we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{\theta}_1}\right) - \frac{\partial L}{\partial \theta_1} = 0,\ \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{\theta}_2}\right) - \frac{\partial L}{\partial \theta_2} = 0 \tag{2}$$

Lagrangian Physics and Double Pendulum

Now let's move on to physics, deriving expressions for energy.

$$\text{Position of Ball 1: } x_1 = l_1 \sin\theta_1,\ y_1 = -l_1\cos\theta_1$$
$$\text{Position of Ball 2: } x_2 = x_1 + l_2 \sin\theta_2,\ y_2 = y_1 - l_2\cos\theta_2$$

So,

$$\text{Speed of Ball 1 } v_1 = \frac{\mathrm{d}x_1}{\mathrm{d}t},\ \text{Speed of Ball 2 } v_2 = \frac{\mathrm{d}x_2}{\mathrm{d}t}$$
$$\text{Total } \mathrm{E}_k = \frac{1}{2}m_1(v_1)^2 + \frac{1}{2}m_2(v_2)^2,\ \text{Total } \mathrm{E}_p = m_1 g y_1 + m_2 g y_2$$
$$L = T - V = \mathrm{E}_k - \mathrm{E}_p = \frac{1}{2}m_1(v_1)^2 + \frac{1}{2}m_2(v_2)^2 - m_1 g y_1 - m_2 g y_2$$

Double Pendulum Equation System Derivation

Substitute this Lagrangian L expression back into formula (2) mentioned before,

For example, for $\frac{\partial L}{\partial \theta_1}$, the partial differentiation could be calculated as:

$$\begin{aligned}\frac{\partial L}{\partial \theta_1} = \frac{\partial}{\partial \theta_1}\Bigg[&\frac{1}{2}m_1(l_1^2\cos^2\theta_1\cdot\dot{\theta}_1^2 + l_1^2\sin^2\theta_1\cdot\dot{\theta}_1^2)\\ &+\frac{1}{2}m_2(A+B)\\ &-m_1 g\left(-l_1\ \cos\theta_1\right) - m_2 g\left(-l_1\cos\theta_1 - l_2\cos\theta_2\right)\Bigg]\end{aligned}$$

$$A = l_1^2\cos^2\theta_1\cdot\dot{\theta}_1^2 + l_2^2\cos^2\theta_2\cdot\dot{\theta}_2^2 + 2\,l_1 l_2\cos\theta_1\cos\theta_2\cdot\dot{\theta}_1\dot{\theta}_2,$$

$$B = l_1^2\ \sin^2\theta_1\cdot\dot{\theta}_1^2 + l_2^2\ \sin^2\theta_2\cdot\dot{\theta}_2^2 + 2\,l_1 l_2\ \sin\theta_1\ \sin\theta_2\cdot\dot{\theta}_1\dot{\theta}_2$$

After repeating this procedure 4 times, the equation could be expanded and solved.

Methodology and Experiment

The system of differential equations we use in this project can be written as:
(The derivation process is avaliable in the appendix).

$$\alpha_1 = \frac{\mathrm{d}\omega_1}{\mathrm{d}t} = \frac{-g(2m_1 + m_2)\sin(\theta_1) - m_2 g \sin(\theta_1 - 2\theta_2) - 2\sin(\Delta) m_2(\omega_2^2 l_2 + \omega_1^2 l_1 \cos(\Delta))}{l_1 \cdot [2m_1 + m_2 - m_2 \cos(2\theta_1 - 2\theta_2)]}$$

$$\alpha_2 = \frac{\mathrm{d}\omega_2}{\mathrm{d}t} = \frac{2\sin(\Delta)\left[\omega_1^2 l_1(m_1 + m_2) + g(m_1 + m_2)\cos\theta_1 + w_2^2 l_2 m_2 \cos\Delta\right]}{l_2 \cdot [2m_1 + m_2 - m_2 \cos(2\theta_1 - 2\theta_2)]}$$

Where,

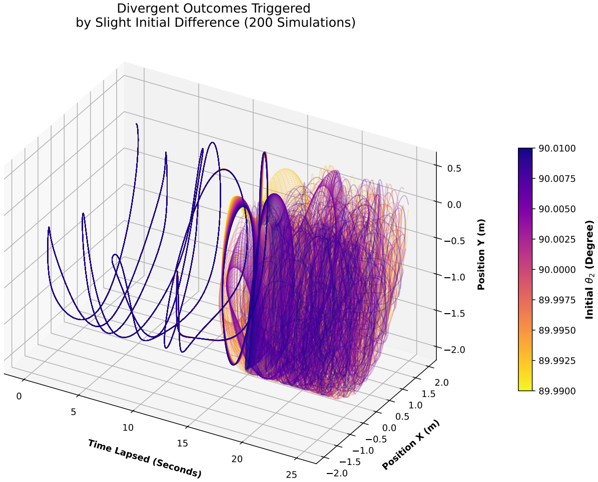
$$\omega_1 = \frac{\mathrm{d}\theta_1}{\mathrm{d}t},\ \omega_2 = \frac{\mathrm{d}\theta_2}{\mathrm{d}t},\ \Delta = \theta_1 - \theta_2$$

In this system, $\theta_\bullet$, $\omega_\bullet$, and $\alpha_\bullet$ respectively represents angular position, angular speed and angular acceleration for each part of the pendulum. $m_\bullet$ is the symbol of the mass of each pendulum, and $l_\bullet$ is the length of each pendulum.

Methodology and Experiment

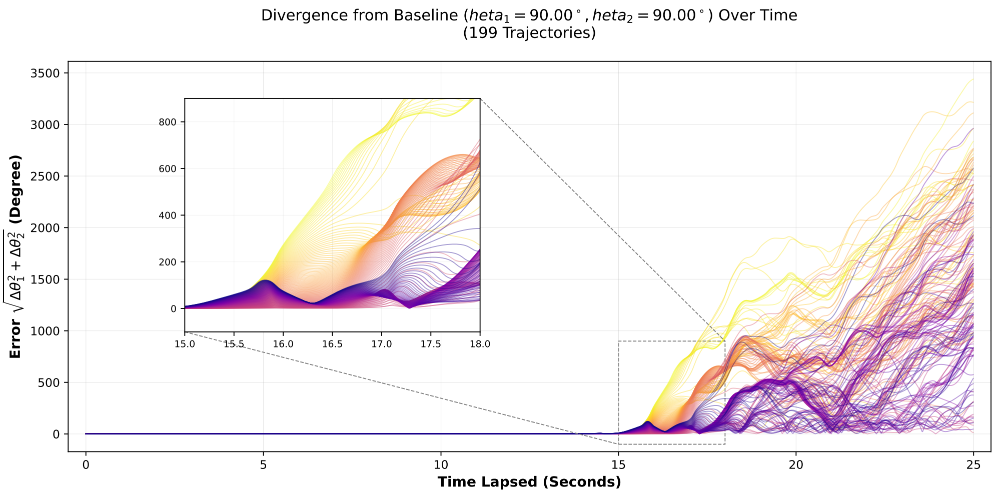
- **Experiment 1:** Simulating double pendulum motions with slight initial difference, observing and discovering the *chaos* in such systems. Visualize experimental results, and desctibe.
- **Experiment 2:** Observing the approach of simulation results under different time step (Δt), and evaluate the efficiency of simulation under different accuarcies.
- **Experiment 3:** Evaluating the pattern of the change of chaoticness of the system under different initial conditions.

Experiment 1 Results



This graph I draw at the left illustrates the position of 2nd pendulum as time goes. In first 15 seconds, the chaoticness wasn't significant, but after that, outcomes become divergent.

Experiment 1 Results

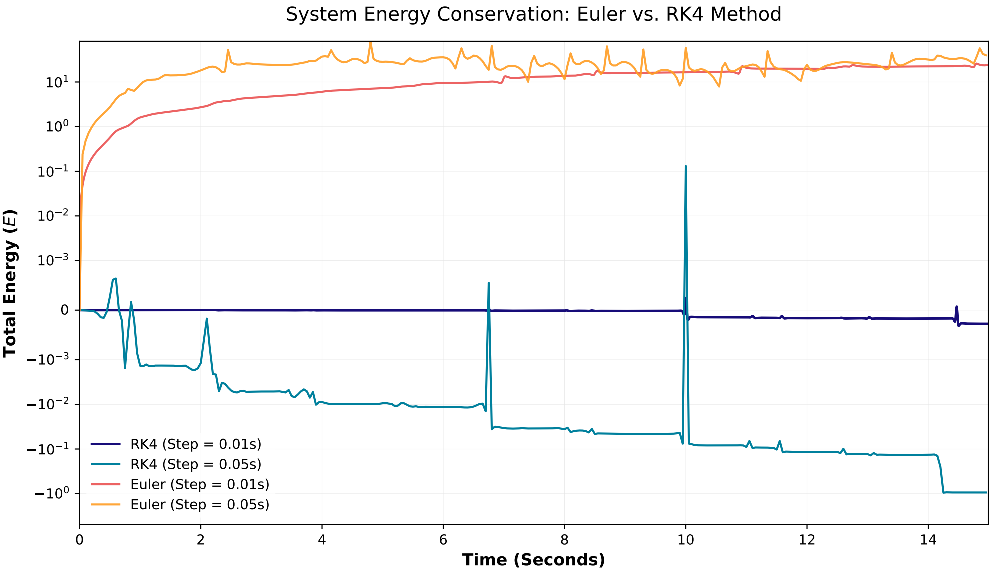


Experiment 2 Results

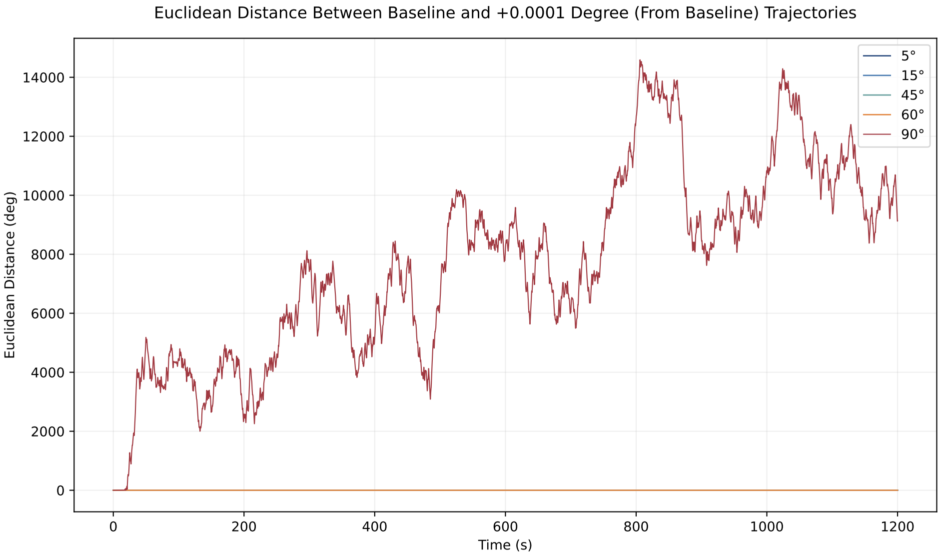
Simulation Results for θ_1 With Different Time Steps

Set	.2s	.1s	.05s	.01s	.005s	.001s	.0005s	.0001s
$\theta_{1,\,\mathrm{RK4},\,1\mathrm{s}}$	-35.61	-35.95	-35.81	-35.81	-35.81	-35.81	-35.81	-35.81
$\theta_{1,\,\mathrm{Euler},\,1\mathrm{s}}$	-69.97	15.51	-15.53	-33.47	-34.77	-35.63	-35.72	-35.80
$\theta_{1,\,\mathrm{RK4},\,5\mathrm{s}}$	-29.87	-35.81	-35.96	-35.94	-35.94	-35.94	-35.94	-35.94
$\theta_{1,\,\mathrm{Euler},\,5\mathrm{s}}$	N/A	45.33	18.28	-68.79	-41.16	-36.94	-36.46	-36.04
$\theta_{1,\,\mathrm{RK4},\,10\mathrm{s}}$	-39.03	6.86	-27.70	-32.57	-32.57	-32.57	-32.57	-32.57
$\theta_{1,\,\mathrm{Euler},\,10\mathrm{s}}$	N/A	-371.64	24.71	181.77	-12.38	-86.36	-88.17	-53.18
$\theta_{1,\,\mathrm{RK4},\,15\mathrm{s}}$	77.44	-14.61	35.17	-40.09	-40.85	-41.44	-41.51	-41.57
$\theta_{1,\,\mathrm{Euler},\,15\mathrm{s}}$	N/A	-78.47	-225.49	-716.50	-92.05	-51.32	31.15	-80.49

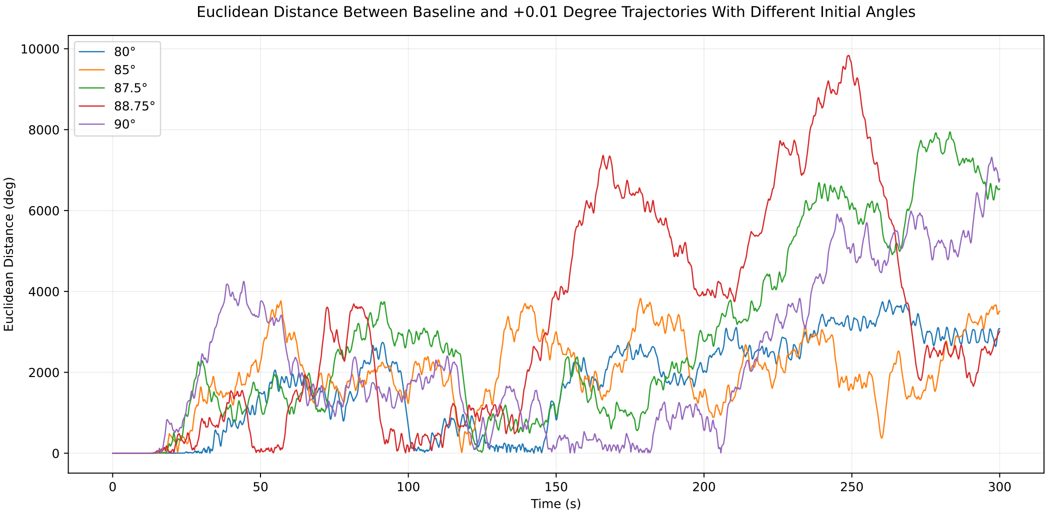
Experiment 2 Results: An Energy View



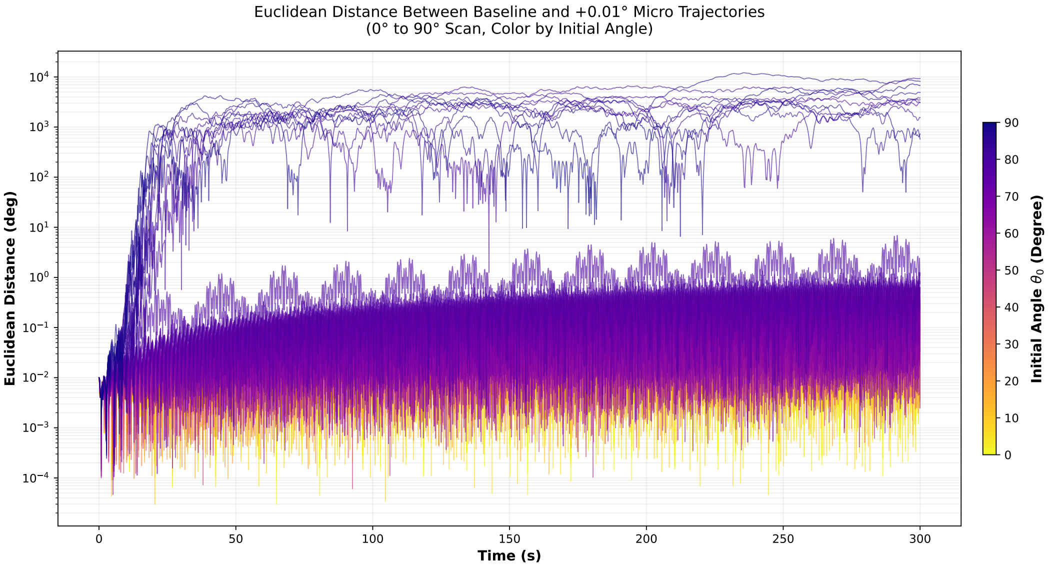
Experiment 3 Results



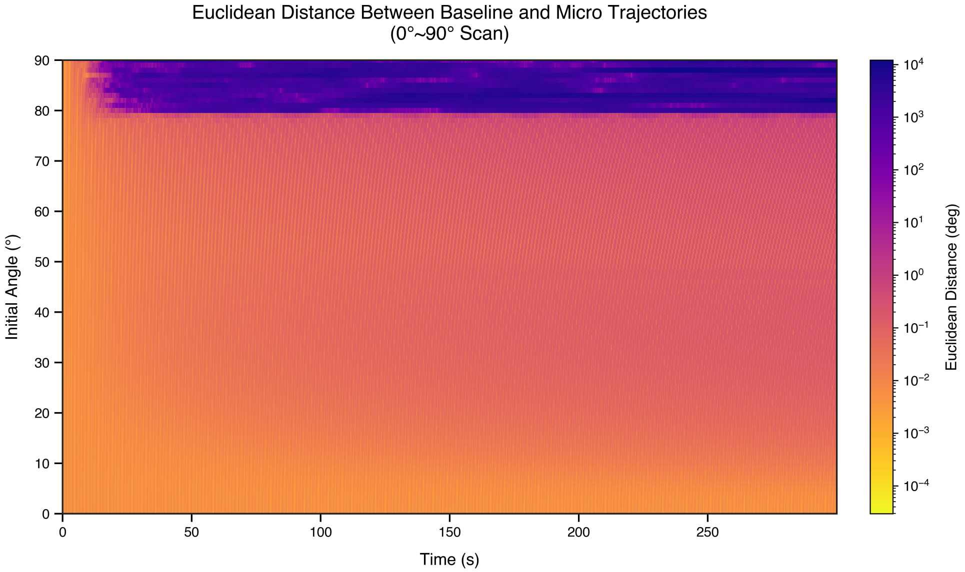
Experiment 3 Results



Experiment 4 Results



Experiment 4 Results



Additional Information

This research is assisted by Artificial Intelligence, but the author promises that AI was not used to create any origional work, all content that is polished or edited by AI is manually validated, the author declares that de did not use AI inappropriately to any extent during this research, and is fully responsible for this research.

Graphs are generated with experimental data, and the copyright is protected. All Code is available at:

`https://github.com/winstonhuangHZ/CALC_EOCPROJ`.